Softwarica
COLLEGE OF IT & E-COMMERCE

in collaboration with

Coventry University

ST7072CEM MACHINE LEARNING · COURSEWORK

Analysis and Prediction of Monsoon Hazards in Nepal Using Machine Learning Approaches

Will the monsoon hurt someone in this district in the next three days?

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BACKGROUND

Every monsoon costs lives and homes

2,561
deaths from 2017 to 2025

8,586
houses destroyed

56,379
families affected

AND IN 2026

The Bhotekoshi flood in Rasuwa killed **more than 1,300** people on 26 August, with thousands missing: about as many as 2021 to 2025 combined.

Deaths per year from floods, landslides, heavy rain and lightning

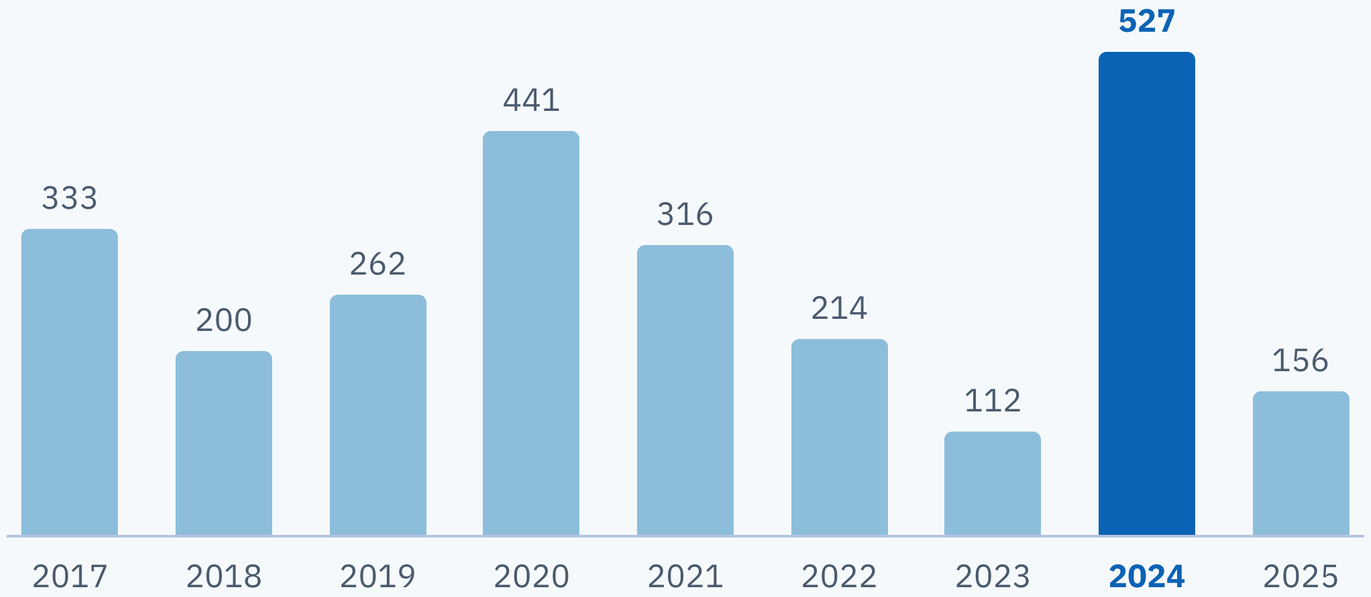


Chart: BIPAD disaster record, 2017 to 2025. 2026 flood: NDRRMA figures, 6 September 2026 (1,341 dead, 5,083 missing)

PROBLEM STATEMENT

Acting early needs a forecast of harm

THE POLICY

Release relief **before** a hazard strikes, on forecast triggers

WHAT IT NEEDS

A forecast of **which district** will be hurt, and **when**

WHAT EXISTS

Maps of **where** hazards can happen, not when

The gap: existing work does not forecast, district by district and day by day, whether the monsoon will cause a death, injury, missing person or destroyed house in Nepal.

Policy: NDRRMA, National Anticipatory Action Framework 2026, Government of Nepal

AIM AND OBJECTIVES

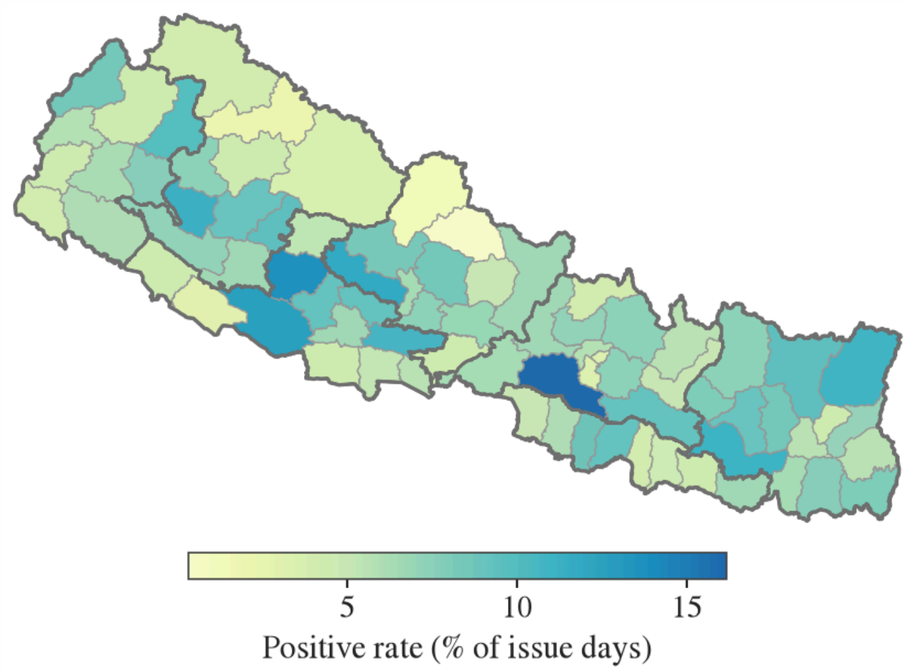
Can machine learning tell when and where monsoon harm will happen?

1. **Cluster** the 77 districts into risk regions with similar harm history and rain climate
2. **Classify** whether a district will suffer a damaging incident in the next three days
3. **Test** whether the regions improve the forecast, and explain what drives the risk

> If the forecast works, it can serve as a trigger for early relief.

DATASET

Built from open records and rain



Share of days followed by harm, by district

16,790

flood, landslide, heavy-rain and lightning records from BIPAD, 2011 to 2026

146,223

district-days, April to October 2017 to 2025, with ERA5 rainfall

6.8%

followed by harm, so the classes are very unbalanced

> Harm rises with rain, from 3.3% of days in the driest tenth to 14.7% in the wettest. Features only use what was known on the day.

METHODS AND EXPERIMENTAL SETUP

Two stages, tested on unseen seasons

STAGE 1

Group districts

K-means, Ward and a Gaussian mixture find 6 risk regions

STAGE 2

Forecast harm

Logistic regression, random forest, gradient boosting, two support vector machines

TEST

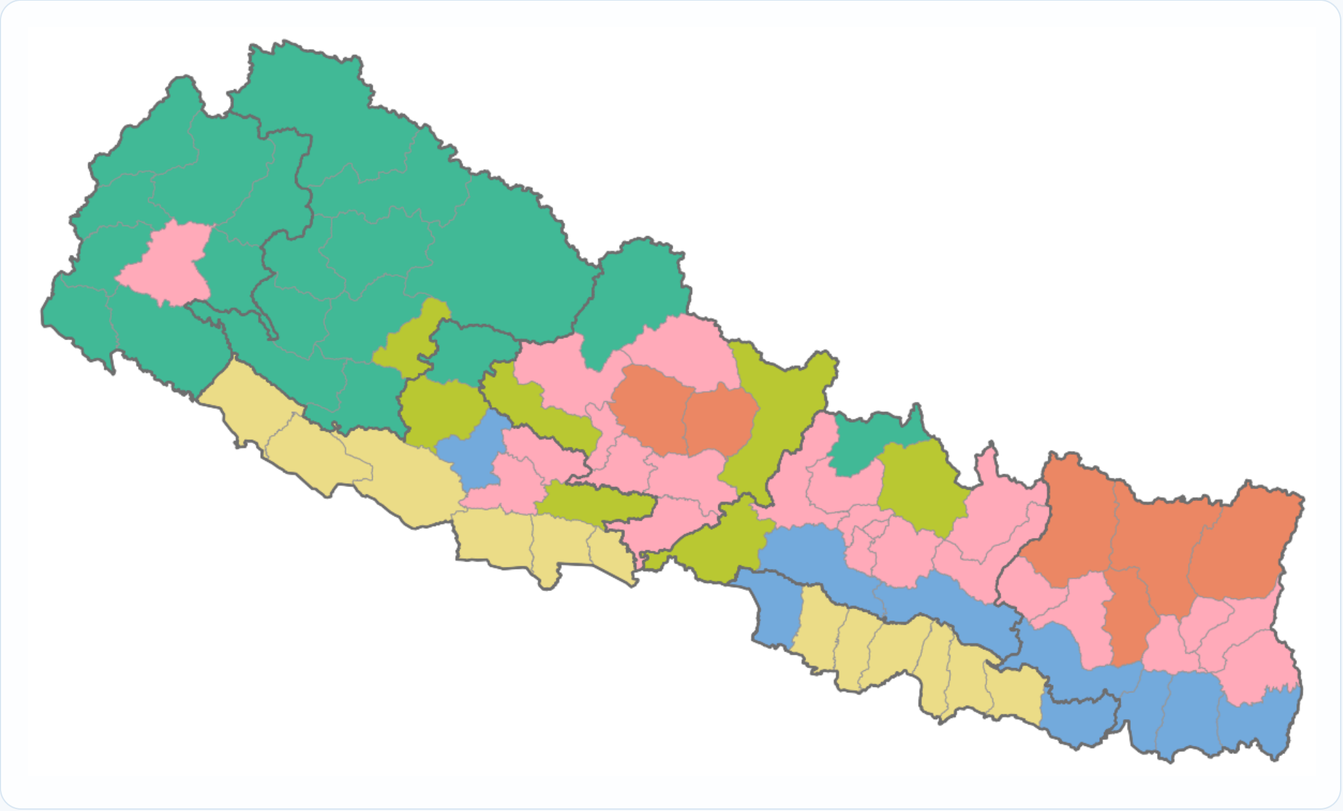
Beat the usual

Against climatology: each district’s normal rate for that month



RESULTS: RISK REGIONS

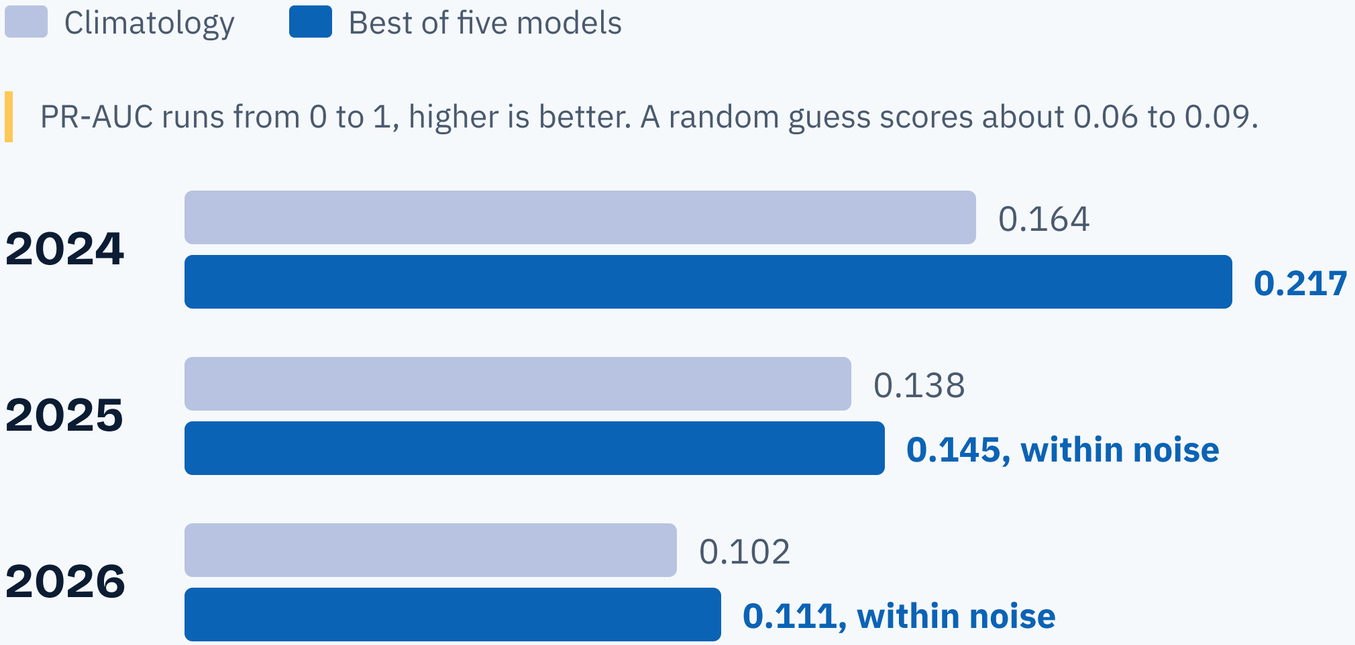
Six regions, each with its own pattern of harm



- **Eastern and inner Terai**
  Most lightning harm · 9 districts
- **Central and western Terai**
  Floods and heavy rain · 12 districts
- **Landslide-prone hills**
  Highest damage rate · 7 districts
- **Wet eastern and central hills**
  July landslide peak · 6 districts
- **Central hill belt**
  Includes Kathmandu valley · 23 districts
- **Dry far west and north-west**
  20 districts

RESULTS: FORECAST

A real gain, but only in one season



+0.026

mean PR-AUC gain over climatology in 2024 and 2025

Logistic regression, gradient boosting and linear SVM

DISCUSSION AND CONCLUSIONS

What each objective found

1 · Risk regions	Six regions, moderately stable, each with its own pattern of harm
2 · Forecast	Beats climatology by about 0.026 PR-AUC, but only in 2024
3 · Do regions help?	No. Knowing the next three days of rain would lift PR-AUC to about 0.21

A harm forecast for Nepal is possible, but its value depends on **better rain forecasts** and on testing it season by season.

WALKTHROUGH

The report and the code

Python and Jupyter notebooks, from raw records to the results in the paper.

GITHUB REPOSITORY
bhimrazy/st7072cem-ml-assignment

01 **The report**
The full paper, in IEEE format

02 **The repository**
How to set it up, and what is where

03 **Data and regions**
Data checks, then notebooks 01 to 03, run live

04 **Models and results**
Notebooks 04 and 05, with their saved outputs

LIMITATIONS

BIPAD records reported harm, so reporting bias shapes the label

Lightning is the hardest hazard to forecast from daily rain

A live system would need rain forecasts, not observed rain

Thank you

github.com/bhimrazy/st7072cem-ml-assignment

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